

PERSON COUNTY AP, NC, USA

WMO: 722193

Lat: 36.285N

Lon: 78.984W

Elev: 609

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03722

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	20.9	-1.8	5.1	27.5	2.9	6.5	30.1	19.9	55.1	18.1	54.4	2.4	310	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	93.0	73.6	90.6	73.4	88.4	72.9	77.5	87.8	76.4	85.7	75.4	83.8	5.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.0	134.5	80.1	73.3	126.8	78.9	72.8	124.3	78.4	41.3	87.4	40.1	86.2	39.2	83.1	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	14.4	12.2	9.2	95.8	6.4	3.5	4.6	98.3	0.8	100.4	-2.8	102.4	-7.4	104.9		
(5)			8.0	78.8	6.0	1.2	3.7	79.7	0.1	80.4	-3.2	81.1	-7.6	82.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.7	39.6	42.2	49.2	58.3	66.6	74.1	77.1	75.6	70.0	59.0	48.0	43.7	
	DBStd	15.45	10.71	9.62	9.90	8.24	6.92	5.22	4.30	4.83	5.92	8.22	8.71	8.85	
	HDD50	1169	353	256	141	21	0	0	0	0	0	17	141	240	
	HDD65	3650	788	639	496	231	70	3	0	1	25	221	514	662	
	CDD50	4348	30	38	116	269	514	724	841	794	600	298	80	44	
	CDD65	1352	0	1	6	29	119	277	376	329	175	36	3	1	
	CDH74	12074	1	5	78	354	1027	2592	3457	2866	1351	314	26	3	
	CDH80	4509	0	0	8	69	272	1012	1462	1167	452	66	1	0	
Wind	WSAvg	4.1	4.7	5.2	5.6	5.7	4.2	3.5	3.1	2.6	3.3	3.6	3.8	4.1	
Precipitation	PrecAvg	45.0	3.5	2.8	4.4	3.5	3.6	4.1	4.5	4.2	4.4	3.4	3.1	3.4	
	PrecMax	60.0	7.4	5.6	8.1	6.4	7.7	12.0	6.7	9.6	16.9	10.6	8.8	7.7	
	PrecMin	32.2	1.1	1.2	1.0	1.1	0.9	1.6	2.6	1.4	0.7	0.0	0.4	1.1	
	PrecStd	6.6	1.5	1.1	1.9	1.6	1.6	2.5	1.2	1.8	3.7	2.4	2.1	1.6	
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.3	73.1	81.2	86.3	90.2	96.8	97.1	97.4	91.3	86.3	78.6	72.2
MCWB			59.9	59.4	61.4	63.5	70.0	72.4	74.8	74.4	70.0	70.3	62.7	63.1	
2%		DB	65.9	69.7	75.2	81.6	86.2	91.2	92.7	92.6	88.3	81.2	72.3	67.8	
		MCWB	58.0	56.6	58.4	62.3	69.3	73.1	74.7	74.1	70.0	67.5	59.8	61.1	
5%		DB	62.8	64.0	72.0	78.9	82.2	89.6	90.4	89.9	85.6	77.3	68.4	63.6	
		MCWB	55.6	54.4	57.1	61.4	68.1	72.9	74.5	73.8	70.7	66.1	59.4	58.5	
10%		DB	57.0	60.7	66.3	73.4	80.7	86.0	88.1	86.4	81.8	73.1	64.0	60.6	
		MCWB	49.7	50.7	55.0	60.5	67.0	72.2	74.2	73.1	69.3	64.5	56.4	54.0	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures		0.4%	WB	63.3	63.1	65.1	71.2	73.4	78.0	78.9	79.2	76.0	74.3	66.8	66.2
			MCDB	66.0	69.1	75.2	79.0	83.1	86.8	88.3	88.2	82.9	78.8	70.7	69.2
	2%	WB	60.1	60.4	62.4	67.1	71.7	76.3	77.7	77.4	74.3	71.7	64.2	63.0	
		MCDB	64.7	65.8	69.6	75.5	80.7	85.9	87.9	88.1	81.3	76.5	68.0	65.0	
	5%	WB	56.6	56.9	60.2	64.7	70.4	75.0	76.6	76.0	72.9	69.5	61.6	60.0	
		MCDB	62.1	64.3	67.9	72.8	78.4	84.4	86.2	85.0	79.8	75.0	67.1	63.8	
	10%	WB	51.5	51.8	57.5	62.8	69.0	73.7	75.6	74.8	71.7	66.5	57.5	54.6	
		MCDB	55.7	58.4	64.6	70.3	76.4	82.5	84.2	82.7	78.1	72.4	63.5	58.6	
	Mean Daily Temperature Range	5% DB	MDBR	19.6	21.2	22.3	23.8	21.3	20.8	19.9	19.5	19.8	22.1	22.6	19.7
			MCDBR	24.2	26.8	26.4	28.0	24.2	23.9	23.0	23.6	25.4	25.5	27.3	23.2
MCWBR			18.0	17.4	13.8	12.5	10.2	8.3	7.8	8.0	9.4	13.0	16.3	17.6	
5% WB		MCDBR	21.8	23.6	22.5	21.6	19.7	21.0	19.9	19.6	18.5	19.6	20.6	19.1	
		MCWBR	18.6	17.7	14.4	11.6	9.0	8.3	7.3	7.8	8.3	11.5	15.4	17.0	
Clear-Sky Solar Irradiance	taub		0.305	0.323	0.355	0.396	0.441	0.487	0.524	0.522	0.412	0.370	0.324	0.309	
	taud		2.513	2.457	2.411	2.297	2.198	2.113	2.029	2.015	2.322	2.422	2.510	2.536	
	Ebn at Noon		284	291	290	281	269	256	245	242	267	270	274	275	
	Edh at Noon		25	30	35	41	46	50	54	54	37	31	25	23	
All-Sky Solar Radiation	RadAvg		798	1017	1341	1688	1871	2021	1893	1720	1421	1153	877	688	
	RadStd		55	120	96	131	150	143	117	88	129	145	69	59	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (56 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+259	+173	

Nomenclature: See separate page